

Kushagra Lal

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Sector 11-T5, 43/E, Reliance Greens Motikhavdi, Jamnagar – 361142, Gujarat

OBJECTIVE

Aspiring Automobile Engineering student specialising in Automotive Electronics, with a strong interest in Powertrain systems, Electric and Hybrid Vehicle Technology, and Engine Harness & Wiring. Keen to apply academic knowledge and hands-on skills to contribute meaningfully in the automotive industry.

EDUCATION

SRM Institute of Science and Technology	2021 – 2025
<i>Bachelor of Engineering – Automobile Engineering (Automotive Electronics)</i>	<i>CGPA: 6.18/10</i>
Nalanda Academy, Kota	2021
<i>12th Grade</i>	<i>71%</i>
KDAV (Kokilaben Dhirubhai Ambani Reliance Foundation School), Jamnagar	2019
<i>10th Grade</i>	<i>75%</i>

RESEARCH INTERESTS

Powertrain systems, Hybridization of vehicles, Electric Vehicle Battery Management Systems (BMS), Hybrid Vehicle Energy Management System (EMS), E-Mobility, Hydrogen Fuel Cells Technology, EV and HEV technologies, Vehicle Electrical management and diagnosis, Automotive simulations.

PROJECTS

- Second-Hand Car Price Generator** | *C Language, Data Analysis* Aug 2021 – Dec 2021
- Compiled Excel datasets for 40+ automakers covering all models sold in India since market debut.
 - Co-developed the pricing algorithm in C, leveraging data analysis from CSV files to generate viable resale values.
 - Model successfully produced accurate second-hand vehicle valuations.
 - [GitHub Repository](#)
- Quick Up-shifter for Formula student Race cars.** | *Electrical Modification, Model Installation* Jan 2026 – May 2026
- Wrote the Arduino IDE code for the quick-shifter mechanism making sure no false triggers take place, Engine cut-off time set exactly at 0.65 - 0.85 ms
 - Developed full scale working model using Load cell, Analog to Digital converter, Buck converter, Arduino R3, Relay.
 - Model successfully helped the vehicle attain quick shift.
 - [GitHub Repository](#)

TECHNICAL SKILLS

Languages: Python, MATLAB, Arduino IDE, ESP-IDF
Software & Tools: SolidWorks, AutoCAD, Microsoft Office (Word, Powerpoint, excel), Google sheets, Adobe Photoshop, ANSYS, MATLAB, KiCAD, Blender
Operating Systems: Windows 10/11, MacOS
Hardware & Electronics: PCB Design & Repair, Automotive Electrical Wiring Management & Inspection, Soldering, Vehicle Lighting Modifications, Automobile Powertrain,
Other Skills: 3D Mechanical Designing & Parts Simulation, Front-End Coding, Content Writing, Data Entry.

CERTIFICATIONS

Data Science & Analytics • Data Wrangling • JavaScript • Tableau for Beginners • Tableau Desktop • Data Visualization using Excel • Web Scraping

[View Certificates on GitHub](#)

All obtained: May 2025

EXTRA-CURRICULAR ACTIVITIES

Powertrain Head

Team 1.618 (Team Golden Ratio) June 2025 - August 2026

LANGUAGES

English • Hindi